

Regularisation – Lesson overview

Regularisation is an important element in data science, used to create models that generalise well and can **predict accurately on unseen data**. It's a critical defence against **overfitting**, ensuring models remain **adaptable** and **reliable**.

In this lesson, we'll unravel the mechanics and **benefits of regularisation methods**, including **ridge** and **LASSO** regression. We'll delve into the intricacies of **data scaling**, **overfitting**, and the strategic application of **regularisation** to **enhance model performance** in real-world data science scenarios.

 Video	 Examples
 Knowledge questions	 Exercise
 Multiple choice questions	



Learning objectives

- Explain the concept of overfitting and the importance of regularisation in machine learning.
- Differentiate between L1 and L2 regularisation and understand their applications.
- Utilise regularisation techniques in building more accurate and robust machine learning models.
- Implement data scaling techniques to improve model performance.
- Apply ridge and LASSO regression methods to prevent overfitting.

